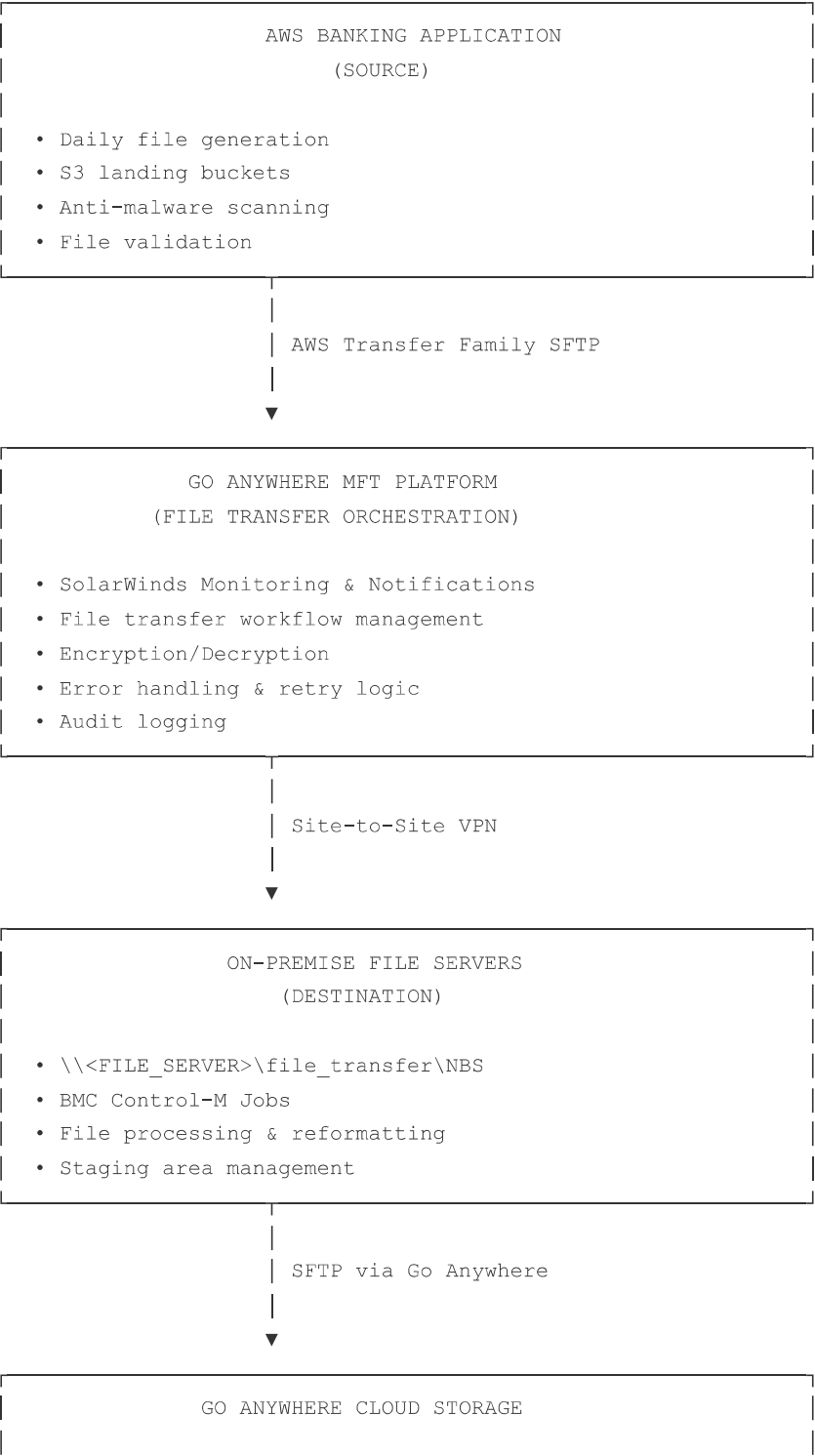


Test Location Matrix: AWS to On-Premise to Go Anywhere

File Transfer Testing - Clear Test Location Assignments



Architecture Overview



- Intermediate storage
- SolarWinds monitoring

Final SFTP Transfer

UNIT4 AGRESSO CLOUD
(FINAL DESTINATION)

- <AGRESSO_SFTP_HOST>
- \\Server location\uk_xxx_prod\$\Data Import



Test Location Decision Matrix

Test Scenario	Primary Test Location	Secondary Location	Monitoring Tool	Who Performs Test
1. File Generation in AWS	✓ AWS Side	-	CloudWatch, Lambda Logs	AWS Team
2. File Upload to MFT	✓ AWS Side	Go Anywhere Platform	Go Anywhere Logs, SolarWinds	AWS Team + MFT Team
3. Anti-Malware Scanning	✓ AWS Side	Go Anywhere (if MFT has AV)	CloudWatch, SolarWinds	AWS Team
4. File Encryption	✓ Go Anywhere Platform	AWS (if done pre-transfer)	Go Anywhere Logs, SolarWinds	MFT Team
5. File Transfer to On-Prem	✓ Go Anywhere Platform	On-Premise (reception)	SolarWinds, BMC Control-M	MFT Team + On-Prem Team
6. File Decryption	✓ On-Premise Side	Go Anywhere (if done at MFT)	BMC Control-M, Windows Event Logs	On-Prem Team
7. Virus Scanning	✓ AWS Side	-	On-Premise Antivirus, Control-M	On-Prem Team
8. File Reception Validation	✓ On-Premise Side	-	BMC Control-M, PowerShell Scripts	On-Prem Team
9. BMC Control-M Processing	✓ On-Premise Side	-	BMC Control-M Dashboard	On-Prem Team
10. File Reformatting	✓ On-Premise Side	-	BMC Control-M Jobs	On-Prem Team

Test Scenario	Primary Test Location	Secondary Location	Monitoring Tool	Who Performs Test
11. SFTP to Go Anywhere Cloud	✔ On-Premise Side	Go Anywhere Cloud	SolarWinds, Control-M	On-Prem Team + MFT Team
12. Go Anywhere Cloud Storage	✔ Go Anywhere Cloud	-	SolarWinds	MFT Team
13. Final Transfer to Agresso	✔ Go Anywhere Cloud	Agresso (reception)	SolarWinds, Agresso Logs	MFT Team + Agresso Team
14. MFT Agent/Platform Down	✔ Go Anywhere Platform	AWS & On-Premise	SolarWinds Alerts	MFT Team
15. Internet/VPN Down	✔ Both AWS & On-Premise	Network Infrastructure	AWS CloudWatch, SolarWinds	Network Team
16. File Recovery from Backup	✔ All Three	Depends on recovery source	Backup software, S3 Versioning	Respective Teams



Detailed Test Breakdown by Location



Tests to Perform on AWS Side

Test Group A: File Generation & Initial Validation

Test ID	Test Name	What to Test	Success Criteria	Monitoring
AWS-001	Daily file generation	Banking app generates files in S3 landing bucket	Files present with correct naming convention	CloudWatch Events, S3 Event Notifications
AWS-002	File naming convention	Files follow pattern: <FileType>_YYYYMMDD.<ext>	All files match naming standard	Lambda validation function
AWS-003	File completeness	File size > 0 bytes, expected columns present	Size check passes, schema validation passes	Lambda metadata check
AWS-004	Multiple file types	.xlsx, .csv, .html files generated correctly	Each format parses successfully	Lambda parser tests
AWS-005	Timestamp accuracy	File creation timestamp matches expected schedule	Files created within ±5 min of schedule	CloudWatch metric

Test Group B: Security & Anti-Malware

Test ID	Test Name	What to Test	Success Criteria	Monitoring
AWS-006	Clean file scanning	Clean files pass AV scan and continue transfer	File remains in landing bucket	Lambda AV function logs
AWS-007	Malware detection (EICAR)	EICAR test file triggers quarantine	File moved to quarantine bucket	CloudWatch alarm, SNS alert
AWS-008	Quarantine isolation	Infected files isolated and not transferred	No infected files reach MFT	S3 bucket policies enforce isolation
AWS-009	Scan performance	Scan completes in < 5 sec for 10MB file	95th percentile < 5 sec	CloudWatch metrics
AWS-010	SNS alert delivery	Malware alert delivered within 30 sec	Alert received by ops team	SNS delivery logs

Test Group C: File Integrity & Validation

Test ID	Test Name	What to Test	Success Criteria	Monitoring
AWS-011	MD5 checksum calculation	Checksum calculated and stored as metadata	Metadata present on S3 object	S3 object metadata
AWS-012	File size validation	Size matches expected range	File within ±20% of average size	Lambda validation
AWS-013	Format validation (XLSX)	Excel files parse without errors	openpyxl successfully reads file	Lambda logs
AWS-014	Format validation (CSV)	CSV files have correct delimiter and encoding	csv.reader parses successfully	Lambda logs
AWS-015	Encoding verification	Files in UTF-8 encoding	Character set validation passes	Lambda validation

Test Group D: File Transfer to MFT

Test ID	Test Name	What to Test	Success Criteria	Monitoring
AWS-016	MFT connection establishment	AWS can connect to Go Anywhere MFT	Connection established < 5 sec	CloudWatch Logs
AWS-017	Authentication success	SSH key or credentials validated	Auth succeeds	Transfer logs
AWS-018	File upload initiation	File transfer starts successfully	Transfer begins	CloudWatch Events
AWS-019	Upload bandwidth	Upload speed meets SLA (e.g., > 10 Mbps)	Speed measured and logged	Custom metric

Test ID	Test Name	What to Test	Success Criteria	Monitoring
AWS-020	Concurrent uploads	Multiple files upload simultaneously	No failures under load	Load test results

Tests to Perform on Go Anywhere MFT Platform

Test Group E: File Reception & Orchestration

Test ID	Test Name	What to Test	Success Criteria	Monitoring
MFT-001	File reception from AWS	MFT receives files from AWS successfully	File present in MFT landing zone	Go Anywhere logs, SolarWinds dashboard
MFT-002	File integrity check	MFT validates MD5 checksum	Checksum matches source	Go Anywhere workflow logs
MFT-003	SolarWinds notification	Alert sent when file received	Notification delivered < 1 min	SolarWinds alert log
MFT-004	Workflow triggering	File reception triggers processing workflow	Workflow executes automatically	Go Anywhere workflow status
MFT-005	File queuing	Files queue properly during high volume	No files dropped	Go Anywhere queue metrics

Test Group F: Encryption & Security

Test ID	Test Name	What to Test	Success Criteria	Monitoring
MFT-006	Encryption (if applicable)	File encrypted before transfer to on-prem	Encrypted file created	Go Anywhere logs
MFT-007	Encryption key management	Correct keys used for encryption	No key errors	Go Anywhere security logs
MFT-008	TLS/SSL validation	Transfer uses TLS 1.2+	SSL certificate valid	Network packet capture
MFT-009	Access control	Only authorized systems can retrieve files	Unauthorized access denied	Go Anywhere audit log
MFT-010	File retention policy	Files deleted after successful transfer	Old files cleaned up	Go Anywhere retention logs

Test Group G: Error Handling & Retry Logic

Test ID	Test Name	What to Test	Success Criteria	Monitoring
MFT-011	Network interruption handling	MFT retries after network failure	Transfer resumes after reconnect	Go Anywhere retry logs
MFT-012	Failed transfer retry	Failed transfers retry up to 3 times	Retry count = 3, then alert	SolarWinds alert
MFT-013	Exponential backoff	Retry delay increases: 1min, 2min, 4min	Timing verified in logs	Go Anywhere workflow logs
MFT-014	Dead letter queue	Failed files moved to error queue	File in DLQ after max retries	Go Anywhere error queue
MFT-015	Alert escalation	Critical failures escalate to on-call	PagerDuty/SMS received	SolarWinds escalation log

Test Group H: Transfer to On-Premise

Test ID	Test Name	What to Test	Success Criteria	Monitoring
MFT-016	VPN connectivity check	MFT can reach on-prem via VPN	Ping/traceroute succeeds	SolarWinds network monitor
MFT-017	File push to on-prem	Files transferred to \\<FILE_SERVER>	File present on file server	Go Anywhere transfer log
MFT-018	Transfer completion notification	Notification sent after successful transfer	Alert received < 1 min	SolarWinds notification
MFT-019	Large file handling	100MB+ files transfer without timeout	Transfer completes	Go Anywhere logs
MFT-020	Bandwidth throttling	Transfer doesn't saturate VPN bandwidth	< 80% bandwidth used	Network monitoring

Tests to Perform on On-Premise Side

Test Group I: File Reception & Validation

Test ID	Test Name	What to Test	Success Criteria	Monitoring
PREM-001	File arrival detection	Files arrive in \\<FILE_SERVER>\file_transfer\NBS	File present in directory	PowerShell script, Control-M sensor
PREM-002	File decryption	Encrypted files decrypted successfully	Decrypted file readable	Decryption script logs

Test ID	Test Name	What to Test	Success Criteria	Monitoring
PREM-003	Virus scanning (on-prem AV)	On-premise AV scans incoming files	Scan completes, no malware found	Antivirus logs
PREM-004	File permissions check	Files have correct NTFS permissions	Read/write access for Control-M account	Windows Event Viewer
PREM-005	Disk space monitoring	Adequate disk space for incoming files	> 10 GB free space	SCOM/Monitoring tool

Test Group J: BMC Control-M Processing

Test ID	Test Name	What to Test	Success Criteria	Monitoring
PREM-006	Control-M job triggering	File arrival triggers Control-M job	Job status = "Started"	BMC Control-M dashboard
PREM-007	Staging copy (P_MIS5_Agresso_MI_Staging_Copy)	Files copied to <STAGING_DRIVE>:\AgressoMIFeed\Staging	Files present in staging	Control-M job log
PREM-008	File preparation (P_COPY_WEBSAVE_FILES_TEMP)	Files read and validated	Job completes successfully	Control-M job log
PREM-009	Reformatting (P_Websave_Agresso_Post_Cloud)	Files converted to fixed-width webtrans.dat	webtrans.dat created	Control-M job log, file validation script
PREM-010	Dependency handling	If file 1 succeeds, file 2 processes	Both jobs complete in sequence	Control-M dependency chain

Test Group K: Error Scenarios (On-Premise)

Test ID	Test Name	What to Test	Success Criteria	Monitoring
PREM-011	Missing file detection	Control-M detects if expected file missing	Alert sent within 30 min	Control-M notification
PREM-012	Corrupted file handling	Corrupted file fails validation gracefully	Job fails, error logged, alert sent	Control-M job log
PREM-013	Empty file handling	Empty (0-byte) file triggers warning	Warning logged, processing skipped	Control-M log

Test ID	Test Name	What to Test	Success Criteria	Monitoring
PREM-014	Invalid format handling	Invalid XLSX/CSV triggers error	Error logged, file quarantined	Control-M error handler
PREM-015	Job failure notification	Failed jobs send email/SMS alert	Alert received < 5 min	Email/SMS logs

Test Group L: File Transfer to Go Anywhere Cloud

Test ID	Test Name	What to Test	Success Criteria	Monitoring
PREM-016	SFTP connection to Go Anywhere Cloud	Connection established from on-prem	Auth succeeds	Control-M SFTP log
PREM-017	webtrans.dat upload	Formatted file uploaded successfully	File present in Go Anywhere Cloud	Go Anywhere storage logs
PREM-018	Upload verification	Control-M verifies file uploaded	File size matches, no errors	Control-M verification script
PREM-019	Post-upload cleanup	Processed files archived/deleted	Staging area cleaned up	PowerShell cleanup script log
PREM-020	Audit log update	Transfer logged in audit database	Audit record created	SQL query on audit table

 Tests to Perform on Go Anywhere Cloud Storage

Test Group M: Cloud Storage & Final Transfer

Test ID	Test Name	What to Test	Success Criteria	Monitoring
GAC-001	File reception in cloud storage	webtrans.dat received from on-prem	File present	SolarWinds storage monitor
GAC-002	Storage retention	Files retained for required duration (e.g., 30 days)	Files older than 30 days purged	Go Anywhere retention policy
GAC-003	File integrity validation	MD5/SHA256 checksum validated	Checksum matches	Go Anywhere validation log
GAC-004	Access control	Only Agresso can access files	Unauthorized access blocked	Go Anywhere audit log
GAC-005	SolarWinds monitoring active	SolarWinds monitors file status	Real-time visibility	SolarWinds dashboard

Test Group N: Final Transfer to Agresso

Test ID	Test Name	What to Test	Success Criteria	Monitoring
GAC-006	SFTP to Agresso gateway	Connection to <AGRESSO_SFTP_HOST>	Connection succeeds	Go Anywhere transfer log
GAC-007	Authentication with Agresso	SSH key authentication succeeds	Auth verified	Agresso SFTP logs
GAC-008	File path validation	File delivered to correct path (\\S-UKSBW-APPP06\uk_wbs_prod\$\Data Import)	Path correct	Go Anywhere logs
GAC-009	Transfer confirmation	Agresso acknowledges file receipt	ACK received	Agresso confirmation log
GAC-010	Final notification	Success notification sent to ops team	Email/SMS received	SolarWinds notification log



Tests to Perform at Network/Infrastructure Level (All Locations)

Test Group O: Network & Connectivity

Test ID	Test Name	Test Location	What to Test	Success Criteria
NET-001	VPN connectivity	AWS ↔ On-Premise	Site-to-Site VPN operational	Both tunnels UP
NET-002	VPN failover	AWS ↔ On-Premise	Secondary VPN activates if primary fails	Failover < 2 min
NET-003	DNS resolution	All locations	Hostnames resolve correctly	nslookup succeeds
NET-004	Firewall rules	All locations	Required ports open (22 for SFTP, 443 for HTTPS)	Connection succeeds
NET-005	Bandwidth testing	All locations	Adequate bandwidth for file transfer	> 10 Mbps sustained
NET-006	Latency measurement	All locations	Latency within acceptable range	< 100ms AWS to on-prem
NET-007	Packet loss	All locations	No packet loss during transfer	0% loss
NET-008	MTU optimization	All locations	MTU configured correctly (1500 for VPN)	No fragmentation
NET-009	Internet outage simulation	All locations	Verify detection and failover	Alert within 2 min



SolarWinds Monitoring & Alerting

What SolarWinds Monitors

Component	Metrics Monitored	Alert Threshold	Notification Method
Go Anywhere MFT Platform	File transfer status, success/failure rate	Failure rate > 5%	Email, SMS, PagerDuty
File Transfer Workflows	Workflow execution time, errors	Duration > 10 min OR any error	Email, Slack
Network Connectivity	VPN status, bandwidth utilization	VPN down OR bandwidth > 80%	SMS, PagerDuty
Storage (Go Anywhere Cloud)	Disk usage, file count	Disk > 90% full	Email
SFTP Connections	Connection failures, auth failures	> 3 failures in 5 min	Email, SMS
File Age	Files not processed within SLA	File age > 2 hours	Email, Slack

SolarWinds Alert Examples

✓ SUCCESS NOTIFICATION:

Subject: File Transfer Success - CIF_Customer_Info_20250101.csv

Body:

- File: CIF_Customer_Info_20250101.csv
- Source: AWS Banking App
- Destination: \\<FILE_SERVER>\file_transfer\NBS
- Transfer Time: 45 seconds
- File Size: 12.5 MB
- Status: SUCCESS
- Next Step: BMC Control-M processing

✗ FAILURE NOTIFICATION:

Subject: URGENT - File Transfer Failed - Deposits_20250101.xlsx

Body:

- File: Deposits_20250101.xlsx
- Source: AWS Banking App

- Destination: \\<FILE_SERVER>\file_transfer\NBS
- Error: Connection timeout after 3 retries
- Retry Count: 3
- Last Attempt: 2025-01-01 10:35:22 UTC
- Status: FAILED
- Action Required: Check VPN connectivity and retry manually
- Runbook: <https://wiki.example.com/file-transfer-failure>

WARNING NOTIFICATION:

Subject: Warning - File Transfer Delayed - Daily_MI_20250101.xlsx

Body:

- File: Daily_MI_20250101.xlsx
- Expected Time: 08:00 AM
- Current Time: 08:45 AM
- Delay: 45 minutes
- Status: PENDING
- Possible Cause: Network latency or source file not generated
- Action: Monitor for next 15 min, then investigate



Test Execution Responsibilities

Team 1: AWS Team

- **Responsible for:** Tests AWS-001 through AWS-020
- **Tools:** AWS Console, AWS CLI, Boto3, Python scripts
- **Duration:** Week 1-2

Team 2: MFT Team (Go Anywhere Specialists)

- **Responsible for:** Tests MFT-001 through MFT-020, GAC-001 through GAC-010
- **Tools:** Go Anywhere web console, SolarWinds, API scripts
- **Duration:** Week 2-4

Team 3: On-Premise Team

- **Responsible for:** Tests PREM-001 through PREM-020
- **Tools:** BMC Control-M, PowerShell, Windows Event Viewer, SQL Server
- **Duration:** Week 3-5

Team 4: Network Team

- **Responsible for:** Tests NET-001 through NET-010
- **Tools:** AWS Console, VPN monitoring, Wireshark, ping/traceroute
- **Duration:** Ongoing (parallel with other tests)



Summary: Where to Test What

Test Category	AWS	Go Anywhere MFT	On-Premise	Go Anywhere Cloud
File Generation	✓	✗	✗	✗
Anti-Malware Scanning	✓	⚠ (optional)	✓	✗
File Encryption	⚠ (optional)	✓	⚠ (decryption)	✗
File Transfer (AWS→On-Prem)	✓ (send)	✓ (orchestrate)	✓ (receive)	✗
Virus Scanning (On-Prem)	✗	✗	✓	✗
BMC Control-M Processing	✗	✗	✓	✗
File Reformatting	✗	✗	✓	✗
SFTP to Go Anywhere Cloud	✗	✗	✓ (send)	✓ (receive)
Final Transfer to Agresso	✗	✓	✗	✓
SolarWinds Monitoring	⚠ (integrated)	✓	⚠ (integrated)	✓
Error Handling & Retry	✓	✓	✓	✓
Backup & Recovery	✓	✓	✓	✓

Legend:

✓ = Primary responsibility ⚠ = Secondary/optional involvement ✗ = Not applicable



Contact Information

Team	Contact	Responsibilities
AWS Team	aws-platform@example.com	File generation, AWS Transfer Family, Lambda functions

Team	Contact	Responsibilities
MFT Team	mft-team@example.com	Go Anywhere MFT, SolarWinds monitoring, file orchestration
On-Premise Team	on-prem-ops@example.com	File reception, BMC Control-M, file processing
Network Team	network-ops@example.com	VPN, Direct Connect, firewall, connectivity
Agresso Team	agresso-support@example.com	Final file reception, Agresso SFTP gateway

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This document clarifies the testing locations for the AWS-to-On-Premise file transfer architecture with Go Anywhere MFT and SolarWinds monitoring.